Colter Snyder

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Hiring Team
Google

Re: Software Engineer III

Dear Hiring Team,

I am a software engineer at Innovative Scientific Solutions, where I build simulation platforms that model nuclear weapons effects for defense agencies. The work rewards two habits Google's engineering culture is known for: measuring before optimizing, and treating correctness as a requirement rather than a goal. I would like to bring those habits, and my M.S. in Computer Science from the Colorado School of Mines, to a Software Engineer III role.

Performance is where I have taken the most initiative. Our team needed to benchmark its calculation services at scale, so I spearheaded an internal Python tool, first as a command-line application and then with a GUI front end, that runs any combination of our 13 effect calculations and charts the results. It exposed a tenfold slowdown in how we call an external library, gave me clear graphs to present the problem to leadership, and saved what would likely have been weeks of manual testing with hand-written JSON inputs. I also port legacy physics models from Fortran and Python to Rust without changing their numerical behavior, and I build the Rust and Python back-end services, Docker deployments, and Angular front ends around them. On our small agile team I lead development of individual features from requirements and design documents through implementation and automated tests, and I review teammates' merge requests regularly, since every change requires a separate reviewer and merger.

Debugging and test engineering have been constants in my work. As an intern at Tyler Technologies I diagnosed a 2,000-test suite and restored about 1,500 tests to passing, so the team could rely on it again. Data structures and algorithms are familiar ground as well: I co-taught CSCI 220, Data Structures and Algorithms in C++, to 120 students, and I taught another section of 90. Explaining an algorithm clearly enough for a room of undergraduates to implement it has made me a better reviewer and a clearer writer of design documents.

I am most energized by problems where the details matter, and I would welcome the chance to bring that to a team at Google. I live in Lakewood and would be glad to work from Google's Boulder office or remotely. Thank you for your time and consideration.

Sincerely,

Colter Snyder